

2022-Ph-H1-Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

A cyclist travels 20 m in 4.0 s. What is their average speed?

Answer: A (5.0 m s⁻¹)

Solution:

$$v = \frac{d}{t} = \frac{20}{4.0} = 5.0 \text{ ms}^{-1}$$

Guidance for Students:

This is a direct application of the average speed formula. Make sure to identify the correct quantities (distance and time) and use consistent units.

Revision Tips:

- Use $v = \frac{d}{t}$ for constant or average speed.
- Keep units consistent (e.g. m and s).
- Watch for distractors — some options may reflect misread numbers.